

YOU and BLUE: An Embodied Robotic Agent for Diaphragmatic Breathing Retraining through Closed-Loop Physiological Computing

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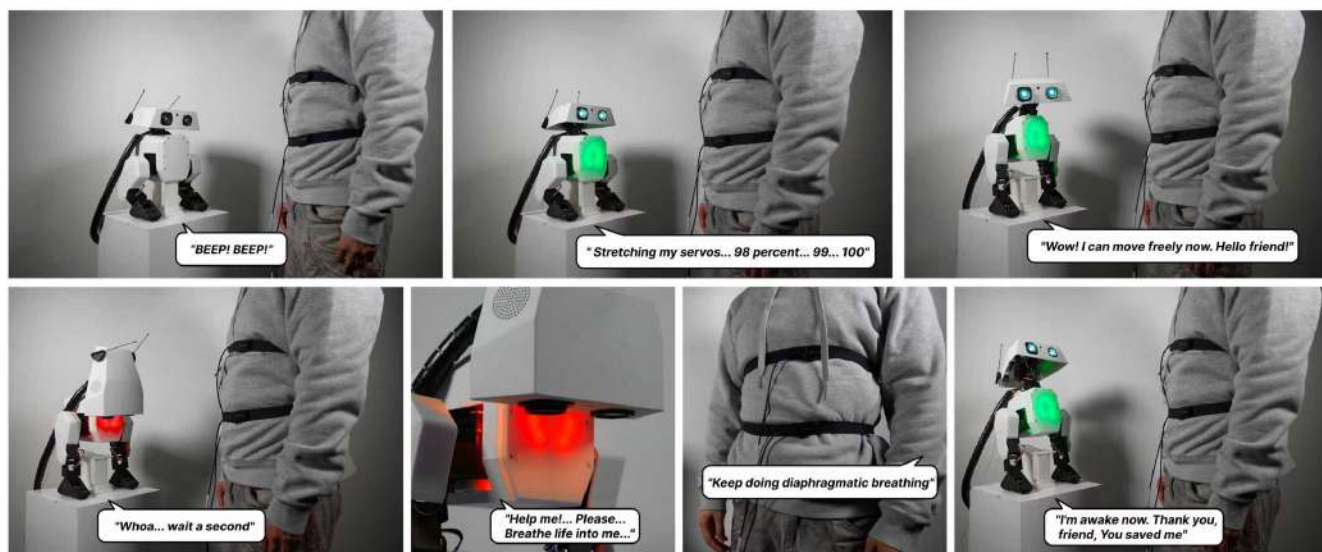


Figure 1: *YOU and BLUE*: experience scenarios. The latest stage of the diaphragmatic breathing retraining is based on a revival-and-rescue framework where *Blue* encourages *You* to keep him alive and explore different interactions together.

Abstract

Diaphragmatic breathing is essential for physiological regulation, yet many people habitually rely on thoracic patterns. We present *YOU and BLUE*, an embodied robotic agent for closed-loop diaphragmatic breathing retraining based on real-time physiological computing. Unlike digital, open-loop biofeedback, our system leverages physical embodiment to provide intuitive respiratory guidance. It integrates dual-channel capacitive respiratory sensing to distinguish thoracic and diaphragmatic movements in real time. A three-stage Finite State Machine model drives the interaction, baseline calibration with visual breathing biofeedback, and paced guidance, while organ-based feedback supports real-time correction. The final stage

is triggered by sustained, correct diaphragmatic breathing. The interaction culminates in a narrative-driven revive-and-rescue scenario in which the robot simulates waking up, standing up, and system collapse, requiring the user to maintain stable diaphragmatic breathing to restore the agent's vitality. This demo illustrates a new paradigm for physiological regulation by combining physiological computing with embodied interaction for diaphragmatic breathing retraining.

CCS Concepts

• **Human-centered computing** → *Interaction paradigms*; **Human computer interaction (HCI)**; **Interaction design**.

Keywords

Physiological computing, Diaphragmatic Breathing Retraining, Robotic Agent, Embodied Interaction

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1 Introduction and Related Work

Breathing is a distinctive physiological process; it is both an automatically regulated metabolic necessity and, at the same time, it is subject to semi-volitional control that can influence affect and arousal [7, 12]. Specifically, diaphragmatic breathing has been empirically shown to engage parasympathetic regulation, reducing anxiety and supporting emotional stability [11, 19]. From the perspectives of ventilation distribution and gas exchange efficiency, some studies suggest that diaphragmatic breathing is more prone to increase ventilation in the lower lung regions, whereas thoracic breathing may yield a more uniform ventilation distribution [13]. However, earlier physiological evidence indicates that these distributional differences are not consistently systematic and can vary with posture, instructed breathing depth, and inter-individual differences in respiratory mechanics [1]. Therefore, in addition to choosing the breathing rate, respiratory mechanics is also an effective approach that can facilitate the transition from a thoracic-dominant mode to a diaphragm-dominant mode in applicable situations.

Physiological computing describes interactive systems that use real-time psychophysiological measures to infer user state and adapt system output in a closed loop [2]. It is often framed as systems that monitor, analyze, and respond to the user's psychophysiological signals, treating physiology as an input channel for interaction [18]. Digital biofeedback can be seen as a common instantiation of physiological computing [2], and has been a long-standing approach [17]. With advances in wearable sensing, contemporary digital platforms can distinguish thoracic and abdominal excursions with relatively high precision and visualize these signals on screens to guide practice [9, 20]. While such systems provide fine-grained physiological data, they remain limited by their screen-based nature, and users need to map abstract visual graphics onto interoceptive and proprioceptive sensations [4], which can increase cognitive load. Moreover, screen-based agents lack physical copresence, often making it difficult to establish a connection and intuitive entrainment needed for long-term engagement.

To address these limitations, previous studies have introduced embodied robotic agents or devices that express breathing rhythms as physical breathing coaches [6, 10, 16, 21]. Prior work suggests that physical embodiment can enhance social presence, trust, and compliance compared to virtual agents [14]. However, many existing robots and devices still rely on open-loop systems that provide breathing guidance but do not verify whether users execute the technique correctly or provide corrective feedback. They also often treat respiration as an undifferentiated signal, and lack sensing that can distinguish thoracic from diaphragmatic patterns, meaning that users may synchronize perfectly with the robot's rhythm while still relying on thoracic breathing, reinforcing the mechanical habit the intervention aims to change. This limitation has also been explicitly noted in research [15] on breathing guidance devices characterized guidance without feedback.

Building on the current research review, where digital systems tend to deliver fine-grained timing control and common relying on

screen-based feedback, embodied systems may deliver more immersive guidance; however, many existing implementations operate in an open-loop system and may not differentiate thoracic from diaphragmatic mechanics. To address this, we present *YOU and BLUE*, an embodied robotic agent for closed-loop diaphragmatic breathing retraining. The system integrates in an open-source robot dual-channel capacitive respiratory sensing across the chest and abdomen to distinguish thoracic from diaphragmatic dominant movements in real time, and structures training through a three stages Finite State Machine (FSM) [8] model: (1) baseline establishment via visual biofeedback, (2) paced guidance through head movements with organ-based feedback from thoracic and abdominal signals, and (3) a narrative revival-and-rescue task in which users restore the robot's vitality by sustaining stable diaphragmatic breathing.

Our contributions are:

1. A closed-loop system based on an embodied robotic agent. We introduce *YOU and BLUE*, which can distinguish thoracic and diaphragmatic breathing patterns through real-time physiological computing. Unlike prior open-loop robotic systems that primarily pace breathing rhythm, our system integrates dual-channel capacitive sensing belts (Elastech) to verify whether users are moving the correct organ and to enable corrective feedback.
2. An interaction paradigm that combines instructional scaffolding with narrative agency. Through a three-stage protocol, we transform repetitive diaphragmatic breathing practice into a meaningful, user-driven experience in which sustained breathing maintains the robot's vitality, linking physiological self-regulation to an engaging affective goal. This coupling reduces monotony while increasing experiential engagement.

2 System Design

We developed and implemented an embodied robotic agent *YOU and BLUE* for closed-loop diaphragmatic breathing retraining through physiological computing. The system uses two capacitive respiratory sensors to continuously sense and visualize users' thoracic breathing and diaphragmatic breathing. Respiration signals are processed in real-time for feature extraction. The robot receives these features and adapts motion, light, and voice to the user's physiological state.

The system supports training and practice to build a stable diaphragmatic breathing pattern. By observing externalized biofeedback cues, users can adjust their breathing strategies promptly. Continued interaction reinforces regulation and supports reflection on bodily states. Fig 2 depicts the system overview.

2.1 Hardware

For the hardware, we built our integrated system upon an *open-source robot*¹ platform and extended it through assembly, testing, and modular modifications. This platform was originally inspired by Disney's BDX prototype [5]. In recent years, alongside advances in embodied intelligence, Disney redesigned the prototype and introduced it as *BLUE*, which was showcased at *NVIDIA GTC 2025*

¹https://github.com/apirrone/Open_Duck_Mini

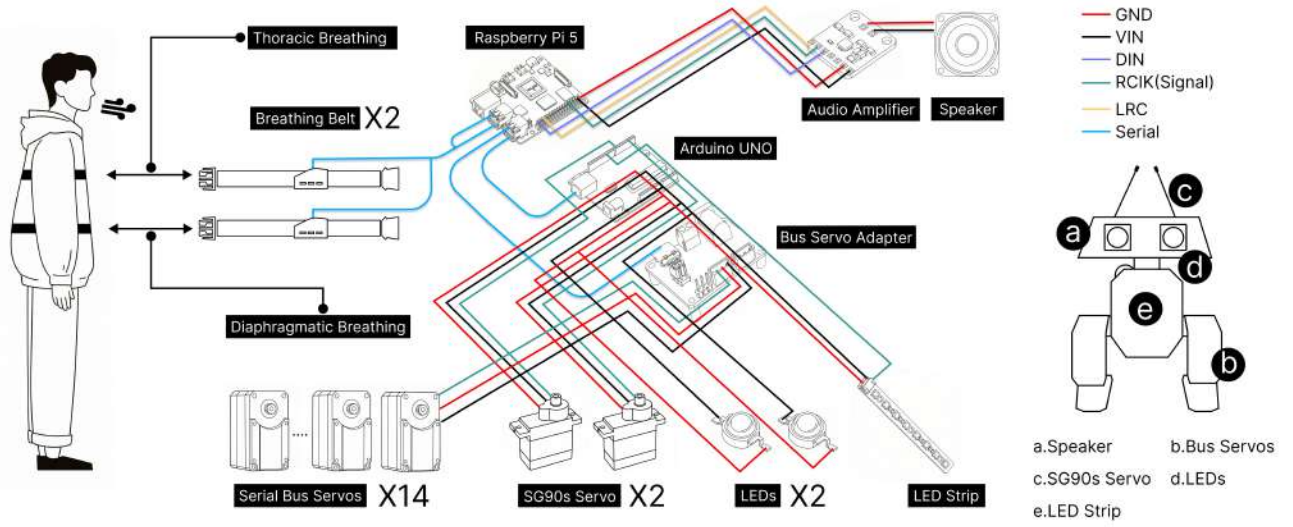


Figure 2: System overview of *YOU and BLUE*. The robotic agent (*Blue*) includes the following modules: a. A speaker for voice prompts; b. Bus servos for body motion; c. SG90S servos for antenna movements; d. LEDs for eye display; e. A WS2812 LED strip on the torso for organ-based breathing feedback. Colored lines indicate power and signal wiring between sensing, computation, and actuation modules.

Developer Conference². Motivated by this development, the open-source community created a low-cost, developer-friendly replica of BLUE, making it suitable for rapid prototyping. Based on this open-source design, we assembled and validated the hardware and expanded capabilities by adding modules and redesigning the system architecture.

The robot body is fabricated with 3D-printed structural components and is actuated by 16 degrees of freedom (DOF) bus servos to enable whole-body motion control. Two SG90S servos rotate the antennas, and two LED beads serve as the eye display unit. For audio output, we use a speaker driven by a MAX98357 audio amplifier module, mounted on the right side of the head. The original open-source design used a Raspberry Pi Zero 2W as the control core. We restructured the control architecture to improve real-time performance and extensibility. We upgraded the host computer to a Raspberry Pi 5 to provide higher compute performance and support more complex multitasking workloads.

In addition, we integrated an Arduino Uno as a microcontroller unit (MCU) to drive the chest-mounted WS2812 LED strip. Compared with the Raspberry Pi, the Arduino provides more reliable support, reducing jitter and blocking caused by the operating system’s scheduling. As a result, LED rendering is more stable and consistent. The system also uses two elastic breathing belts (Elastech), which serve as a basis for monitoring and processing respiration signals from the thoracic and diaphragmatic regions. The system components are shown in Figure 3.

2.2 Software Architecture and Embodied Interaction Design

To enable reliable, real-time closed-loop biofeedback and embodied interaction, we developed a hybrid computing software architecture for *YOU and BLUE*. The system architecture addresses two key challenges. First, breathing belt signals are inherently noisy and exhibit substantial inter-individual variability; second, the robot must synchronize with multimodal outputs under low latency while maintaining temporal consistency. Most software runs on a Raspberry Pi 5 host in Python 3.9. Timing-critical tasks run on an Arduino Uno to avoid host scheduling jitter. These tasks include WS2812 LED refresh and microcontroller-level I/O. The overall software is organized bottom-up into three layers: the Hardware Abstraction Layer (HAL), the Signal Processing Layer, and the Interaction Behavior Layer.

2.2.1 Hardware Abstraction and Hybrid Communication. We built a unified hardware abstraction layer that provides a consistent interface across all devices. It supports fourteen bus servos, two PWM-driven antenna servos, dual-channel respiration sensors, and a WS2812 LED strip. It hides device-specific communication and exposes a consistent sense-to-act interface to upper layers. The host communicates with the MCU through a USB serial port and is responsible for high-level logic and behavior scheduling. For bus servos, we stream commands at 1 Mbps for smooth multi-DOF control. For Arduino devices, we use a lightweight custom ASCII protocol. For example, A:90,90 sets the antenna angles, and L:255 sets the brightness. This division of labor prevents WS2812 refresh from occupying host CPU resources or blocking the main process. It reduces OS-induced timing jitter and improves output stability. We use non-blocking I/O and multi-threading for sensing and actuation.

²<https://www.nvidia.com/en-us/on-demand/session/gtc25-s72484/>

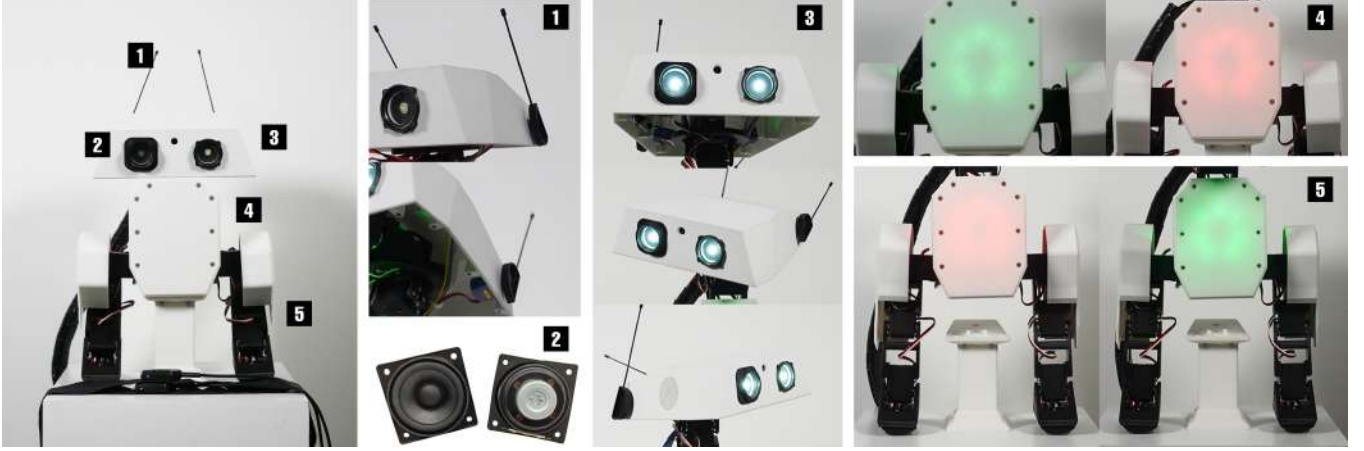


Figure 3: Components of YOU and BLUE: 1. Antennas swinging back and forth; 2. Microphone for voice interaction; 3. Movable head with eyes that brighten and dim; 4. Torso light feedback that turns red for thoracic-dominant mode breathing and green for diaphragmatic breathing; 5. Extendable legs that straighten to lift the body and allow standing.

This design keeps the main control loop stable at 50 Hz with a 20 ms period.

2.2.2 Adaptive Signal Processing Pipeline. A key requirement of a biofeedback system is to map users’ physiological signals into externalized feedback that is stable, interpretable, and aligned with the user’s state. Stretchable capacitive belts are sensitive to placement and tightness. Signals can vary in amplitude and drift across users. We use an adaptive pipeline with dynamic calibration and feature extraction for cross-user consistency. At startup, the software enters a calibration phase called Phase 0, and it samples both thoracic and diaphragmatic channels for 20 seconds. It estimates and locks a personalized min–max range for later normalization and mapping. When mapping respiration to LED brightness, we apply gamma correction to better match human brightness perception and increase discriminability in low-intensity regions. This non-linear mapping produces finer variations of the “breathing light” at low brightness levels, and improving readability and immersion. The system also computes the first-order difference of the thoracic channel in real time. When the Delta chest exceeds a threshold, we infer a shift toward thoracic breathing. The event is sent to the Interaction Behavior Layer to trigger corrective strategies.

2.2.3 Motion Authoring and Affective Audio Tool Chain. We developed a lightweight authoring tool chain for rapid iteration of embodied behaviors and multi-modal content. First, to reduce the cost of designing motions for a 16-DoF robot, we implemented a teaching-by-demonstration motion recording system. Designers can physically manipulate the robot’s limbs while the system records full-joint-angle trajectories at 25 Hz and serializes them to JSON. A playback engine called ActionPlayer parses timestamps to accurately reproduce motions. During celebratory moments, the system randomly selects one action from the recorded action set, ensuring varied robot responses across sessions and enriching users’ experiences.

Second, to give the robotic agent a distinctive, characterful voice, we built an audio post-processing pipeline. After generating base

speech with Edge-TTS, we apply a 400 Hz–4 kHz band-pass filter to simulate a mechanical resonant cavity and add a 12 ms comb-filtering effect to produce a stylized timbre. This processing improves voice recognition and supports consistent characterization across interactions.

2.3 Embodied Interaction Design

To guide users toward retraining and practice of stable diaphragmatic breathing, we designed a three-stage FSM model. Each stage specifies a distinct training objective and a corresponding robot behavior policy. During the experience, participants will wear two elastic motion-detection belts (Elastech): one at the midline between the 4th and 5th intercostal spaces to detect normal (thoracic) breathing [3], and the other at the navel level.

Stage 1: Baseline and Correction. This stage uses visual feedback to capture attention and establish diaphragmatic breathing. *BLUE* remains physically still, while the thoracic lighting mirrors the diaphragmatic channel in real time, allowing users to observe thoracic expansion and contraction directly. The system continuously monitors the thoracic channel. If thoracic breathing increases, the system triggers immediate correction and resets progress. Correction uses red flashing lights and a brief voice prompt. This immediate and contingent negative feedback clearly marks errors and reinforces the target breathing pattern. When the user maintains correct diaphragmatic breathing for 8 seconds, the system transitions to the next stage.

Stage 2: Entrainment and Guidance. In this stage, *BLUE* shifts from following to guiding *YOU*, and uses paced head movements to help users maintain a stable breathing mode. We use the breathing pattern with an inhale of 4 seconds, a hold of 2 seconds, and an exhale of 4 seconds. Head pitch is the primary cue, with raising for inhale and lowering for exhale. To improve engagement, the system uses a soft-tolerance strategy. When users fall out of sync, the robot gives a gentle reminder with head shaking and red flashing lights, and the interaction continues instead of terminating. If the user

follows the paced guidance with correct diaphragmatic breathing for 8 seconds, the system advances to Stage 3.

Stage 3: Narrative Engagement. This stage tackles monotony and declining motivation in biofeedback training. It embeds the exercise within a narrative interaction. The robot first rises from a crouching position to standing, and its antennas and eyes activate. We implement a revival-and-rescue scenario in which the robot first simulates a failure. The head torque is released, the head drops, the antennae droop, and the lights turn off. The user is informed that calm, steady breathing powers *BLUE*. As users sustain stable diaphragmatic breathing, *BLUE* gradually recovers. Lights progressively return, and the head rises through interpolated motion. The sequence ends with recovery and a celebratory performance. By increasing interactivity and playfulness, this design consolidates previously learned diaphragmatic breathing skills while eliciting a deeper user–robot connection.

3 Conclusion

We present *YOU and BLUE*, an embodied robotic agent for diaphragmatic breathing retraining through closed-loop physiological computing. Participants wear two respiration belts at the thoracic and diaphragmatic levels. The system captures and processes dual-channel respiration signals in real time through physiological computing. It renders embodied, multi-modal feedback through lighting, mechanical motion, and voice prompts. During the experience, the system detects thoracic-dominant mode and delivers immediate correction.

YOU and BLUE follows a revival-and-rescue narrative. It acts as both a coach and a companion whose state depends on the user. The quality of diaphragmatic breathing directly shapes *BLUE* state transitions and feedback expression. This framing makes diaphragmatic breathing retraining practice more engaging, effective, and goal-oriented.

GenAI Usage Disclosure

The authors used ChatGPT and DeepL to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- [1] Szonja Decker, Tamás Horváth, Johanna Takács, and Akos Koller. 2024. Body Positions and Physical Activity Levels Modulate the Ratio of Abdominal to Thoracic Breathing and Respiratory Rate in Young Individuals. *Journal of Clinical Medicine* 13, 24 (Dec. 2024), 7825. doi:10.3390/jcm13247825
- [2] Stephen H Fairclough. 2009. Fundamentals of physiological computing. *Interacting with computers* 21, 1-2 (2009), 133–145.
- [3] Robert Gilbert, J Howland Auchincloss Jr, and David Peppi. 1981. Relationship of rib cage and abdomen motion to diaphragm function during quiet breathing. *Chest* 80, 5 (1981), 607–612.
- [4] Oran Goral, Iddo Yehoshua Wald, Amber Maimon, Adi Snir, Yulia Golland, Aviva Goral, and Amir Amedi. 2024. Enhancing interoceptive sensibility through exteroceptive–interoceptive sensory substitution. *Scientific Reports* 14, 1 (June 2024). doi:10.1038/s41598-024-63231-4
- [5] Ruben Grandia, Espen Knoop, Michael A. Hopkins, Georg Wiedebach, Jared Bishop, Steven Pickles, David Müller, and Moritz Bäcker. 2025. Design and Control of a Bipedal Robotic Character. (2025). doi:10.48550/ARXIV.2501.05204
- [6] Venkata Nitin Chakravarthy Gummidela, Dennis R. da Cunha Silva, and Ricardo Gutierrez-Osuna. 2021. Evaluating the Role of Breathing Guidance on Game-Based Interventions for Relaxation Training. *Frontiers in Digital Health* 3 (Dec. 2021). doi:10.3389/fdgth.2021.760268
- [7] Ikuo Homma and Yuri Masaoka. 2008. Breathing rhythms and emotions. *Experimental Physiology* 93, 9 (Aug. 2008), 1011–1021. doi:10.1113/expphysiol.2008.042424
- [8] John E Hopcroft, Rajeev Motwani, and Jeffrey D Ullman. 2001. Introduction to automata theory, languages, and computation. *Acm Sigact News* 32, 1 (2001), 60–65.
- [9] Nobuhisa Ishii, Kazuhide Tomita, Reiko Takeshima, Kenta Kawamura, Yukako Setaka, Ryo Yoshida, and Shigeyuki Imura. 2021. Effects of visual feedback of thoracoabdominal motion on oxygen consumption during hyperventilation - Pilot study. *Journal of Bodywork and Movement Therapies* 28 (Oct. 2021), 317–322. doi:10.1016/j.jbmt.2021.06.007
- [10] Sooyeon Jeong, Sharifa Alghowinem, Laura Aymerich-Franch, Kika Arias, Agata Lapedriza, Rosalind Picard, Hae Won Park, and Cynthia Breazeal. 2020. A Robotic Positive Psychology Coach to Improve College Students' Wellbeing. In *2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 187–194. doi:10.1109/ro-man47096.2020.9223588
- [11] Chan-Young Kwon, Jiyoung Won, and Boram Lee. 2026. The health effects of diaphragmatic breathing: A systematic review. *Complementary Therapies in Medicine* 96 (March 2026), 103317. doi:10.1016/j.ctim.2025.103317
- [12] S. Laborde, M.S. Allen, U. Borges, F. Dosseville, T.J. Hosang, M. Iskara, E. Mosley, C. Salvotti, L. Spolverato, N. Zammit, and F. Javelle. 2022. Effects of voluntary slow breathing on heart rate and heart rate variability: A systematic review and a meta-analysis. *Neuroscience & Biobehavioral Reviews* 138 (July 2022), 104711. doi:10.1016/j.neubiorev.2022.104711
- [13] B. Laufer, S. Krueger-Ziolek, and K. Moeller. 2024. Air Distribution in the Lungs during Abdominal and Chest Breathing. *IFAC-PapersOnLine* 58, 24 (2024), 163–168. doi:10.1016/j.ifacol.2024.11.030
- [14] Kwan Min Lee, Younbo Jung, Jaywoo Kim, and Sang Ryong Kim. 2006. Are physically embodied social agents better than disembodied social agents?: The effects of physical embodiment, tactile interaction, and people's loneliness in human–robot interaction. *International Journal of Human–Computer Studies* 64, 10 (Oct. 2006), 962–973. doi:10.1016/j.ijhcs.2006.05.002
- [15] Kayla Matheus, Ellie Mamantov, Marynel Vázquez, and Brian Scassellati. 2023. Deep Breathing Phase Classification with a Social Robot for Mental Health. In *INTERNATIONAL CONFERENCE ON MULTIMODAL INTERACTION (ICMI '23)*. ACM, 153–162. doi:10.1145/3577190.3614173
- [16] Kayla Matheus, Marynel Vázquez, and Brian Scassellati. 2025. Ommie: The Design and Development of a Social Robot for Anxiety Reduction. *ACM Transactions on Human-Robot Interaction* 14, 2 (Feb. 2025), 1–34. doi:10.1145/3706122
- [17] Eleni Mitsea, Athanasios Drigas, and Charalabos Skianis. 2024. Artificial Intelligence, Immersive Technologies, and Neurotechnologies in Breathing Interventions for Mental and Emotional Health: A Systematic Review. *Electronics* 13, 12 (June 2024), 2253. doi:10.3390/electronics13122253
- [18] Domen Novak, Matjaž Mihelj, and Marko Munih. 2012. A survey of methods for data fusion and system adaptation using autonomic nervous system responses in physiological computing. *Interacting with computers* 24, 3 (2012), 154–172.
- [19] Marcin Sikora, Rafał Mikołajczyk, Olga Łakomy, Jakub Karpiński, Aleksandra Żebrowska, Sabina Kostorz-Nosal, and Dariusz Jastrzębski. 2024. Influence of the breathing pattern on the pulmonary function of endurance-trained athletes. *Scientific Reports* 14, 1 (Jan. 2024). doi:10.1038/s41598-024-51758-5
- [20] Wen-Ruei Tang, Wei Su, Jenn-Jier James Lien, Chao-Chun Chang, Yi-Ting Yen, and Yau-Lin Tseng. 2024. Development of a real-time RGB-D visual feedback-assisted pulmonary rehabilitation system. *Heliyon* 10, 1 (Jan. 2024), e23704. doi:10.1016/j.heliyon.2023.e23704
- [21] Emily Jean Thomas, Kerim Dincel, and Ilona Buchem. 2025. NEFFY – A Social Robot for Multimodal Support of Slow-Paced Breathing Exercises. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, Melbourne, Australia, 1800–1802. doi:10.1109/HRI61500.2025.10974160